

Stefano Fogarollo | Curriculum Vitae

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 sirfoga

Quidquid latine dictum sit, altum sonatur

Education

Enrico Fermi

High school

Liceo Scientifico

Padova, IT

Sep 2009 - Jul 2014

Università degli Studi di Padova

BSc

Computer Science

Padova, IT

Oct 2015 - Jul 2019

TU Dresden

MSc

Computational Modeling and Simulation, Computational Life Sciences

Dresden, DE

Oct 2019 - Sep 2021

University of Innsbruck

PhD

Fast, Non-rigid Image Fusion Using Deep Learning For Treatment Evaluation After Hepatic Thermal Ablation

Innsbruck, AT

Nov 2021 - March 2026

Bachelor thesis

Title: *Implementing IEC 60870-5-104 in XMonitor gateway*

Supervisors: Vittorio De Zuane (ThisOne!), Armir Bujari (Università degli Studi di Padova)

Abstract: Development of a GNU/Linux Qt-based app for collecting data via radio/modbus and broadcast to Monet via IEC 60870-5-104 protocol

MSc thesis

Title: *3D human pose estimation in the wild*

Supervisors: Edoardo Remelli and Sina Honari (EPFL), Bjoern Andres and Stefan Gumhold (TU Dresden)

Abstract: State-of-the-art pose-estimation pipelines assume to have training data in the form of images, camera information, 3D annotations. Such information can only be retrieved in controlled settings, and therefore current pipelines do not work well in the wild (where we don't have access to camera information). The goal of this MSc thesis is to train a neural network only on images and 2D pose information to learn camera information and 3D pose of the human in an unsupervised fashion.

PhD thesis

Title: *Fast, Non-rigid Image Fusion Using Deep Learning For Treatment Evaluation After Hepatic Thermal Ablation*

Supervisors: Prof. Matthias Harders (UIBK), Prof. Reto Bale (MUI)

Abstract: We develop AI-based image analysis methods to improve the treatment evaluation of liver interventions. Key contributions include a contrastive learning framework for robust liver segmentation, a graph-based method for vascular structure alignment, and a novel implicit registration approach. The methods have been retrospectively validated at the Innsbruck University Hospital.

Internships

Zucchetti

Software developer, UML robustness diagrams
Java, JavaScript (ECMAScript), MySQL, UML

Padova, IT

Apr 2018 - Jul 2018

ThisOne!

Software engineer, IOT, Industrial automation
C/C++, Lua, Qt, OpenWrt

Noventana, IT

Feb 2019 - Jun 2019

Amadeus IT Group SA

Software developer, departure control system
C/C++, ORACLE RDBMS

London, UK

Aug 2019 - Sep 2019

Work Experience

Atroos S.r.l.

Full-stack software developer, industry 4.0, predictive maintenance
Node.js, React, MongoDB

Noventana, IT

Oct 2018 - Aug 2019

Scuola Normale Superiore

Full-stack software developer, Cosmology Group
Python 2/3, Apache 2 on CentOS 6+, JupyterHub, Flask, React 15/16

Pisa, IT

Feb 2019 - Feb 2020

Compass Flight Instruments S.r.l.

Embedded software developer, free-flight instruments
C/C++, Embedded Linux, GTK+ 3, BlueZ

Castelfranco Veneto, IT

June 2019 - June 2020

Research Experience

ZIH, TU Dresden

Software developer, HPC
C/C++ 11, Dash, OpenMPI, Fortran95

Dresden, DE

Jan 2020 - November 2020

MPI-CBG

Software developer, Dynamic Load Balancing, Python-C++ Bindings
C/C++ 11, OpenFPM

Dresden, DE

June 2020 - December 2021

BIOTEC, TU Dresden

Software engineer, COVID-19 AASD Modeling
Matlab

Dresden, DE

April 2020 - July 2020

Groupe Weigert, EPFL

Research assistant, Attention Methods for Image Segmentation
Python, Keras

Lausanne, CH

October 2020 - March 2021

LPBS, EPFL

Research assistant, GANs for Microscopy Images
Python, PyTorch

Lausanne, CH

November 2020 - February 2021

CVLAB, EPFL <i>MSc thesis, 3D Human Pose Estimation In The Wild</i> Python, PyTorch	Lausanne, CH <i>February 2021 - September 2021</i>
IGS, University of Innsbruck <i>PhD, AI for Liver SRFA Evaluation</i> Python, PyTorch, C/C++	Innsbruck, AT <i>November 2021 - March 2026</i>
PERK Lab, Queen's <i>Visiting Researcher, AI-based Liver Registration</i> Python, PyTorch	Kingston, ON <i>July 2023 - October 2023</i>
Mori Lab, Nagoya University <i>Visiting Researcher, AI for Treatment Evaluation</i> Python, PyTorch	Nagoya, JP <i>October 2025 - November 2025</i>
IGS, University of Innsbruck <i>PostDoc, Multi-modal Neural Twin</i> Python, PyTorch, C/C++	Innsbruck, AT <i>December 2025 - ongoing</i>

Selected Publications

Towards liver segmentation in the wild via contrastive distillation <i>S. Fogarollo, R. Bale, M. Harders</i> Intuitive Young Investigator Award, IPCAI	IJCARS 2023
Learning Deformable Intra-Patient Liver Registration with GCA <i>S. Fogarollo, Y. Scharll, R. Bale, M. Harders</i>	MICCAI/WBIR 2024
Towards Fast and Accurate Non-rigid Liver Fusion <i>S. Fogarollo, Y. Scharll, R. Bale, M. Harders</i>	MICCAI/WBIR 2024
Implicit Deformable Medical Image Registration with Learnable Kernels <i>S. Fogarollo, G. Laimer, R. Bale, M. Harders</i> Top 9%	MICCAI 2025
Ablate Them All <i>A. Lukes, S. Fogarollo, R. Bale, W. Freysinger</i>	MICCAI/CLIP 2025
AI-Driven Ablative Margin Assessment after RFA of HCC <i>S. Fogarollo, G. Laimer, R. Bale, M. Harders</i>	EURE 2026
Differentiable Manifold Optimization for Ablation Planning <i>P. Floerl, R. Bale, M. Harders, S. Fogarollo</i>	MICCAI/CLIP 2026

Selected Grants

KAT: KI-Gestützte Auswertung von Thermoablationen <i>Approved, not funded</i>	Land Tyrol 2024
hepaTic Neural Twin (TNT) <i>Funded (450K €)</i>	FFG-BRIDGE 2025

Languages

Italiano: Professional working proficiency	<i>mother tongue</i>
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English: Professional working proficiency

B2, FCE

Français: Limited working proficiency

A1, DELF

Deutsch: Limited proficiency

A1, TUDIAS